C - Basics

Type Conversion

- implicit type conversions
  - perform operation with mismatched types
  - assigning values to variables
  - calling functions with type other than specified
  - Examples:
    - int x = 5.0;
    - double y = 5.0/2
    - char c = 50
- Explicit Type Conversions
  - typecasting
  - (type) expression;

Operators

- Assignment: =
- Mathematical:
  - +, -, /, *
  - Mod (remainder): %
  - All can be combined with = to operate and assign
- Increment/decrement: i++, i--
- Comparison: ==, !=, >, <, >-, <=

Logical Operators

- Boolean not: !
- Boolean and: &&
- Boolean or: ||
- Combinations only execute as far as they need to
- Depending on C version, may not have bool
  - Introduced in C99
  - Need to #include <stdbool.h>
- Otherwise, itns act as bools (0 is false)

Control Structures - if

```
if (expression) {
    statement;
}
else {
    statement;
}

if (expression) statement;
```

Ternary Operator

- shorthand for some ifs
- expression -- just chooses which expression based on value of another